

Requirement Analysis Document

Online Book Store System

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1. Project Overview

The Online Book Store is an e-commerce platform that allows customers to browse, search, and purchase books online. This document analyses the business requirements of the platform and translates them into structured functional requirements, non-functional requirements, user stories, and acceptance criteria that can be used for manual testing and QA planning.

2. Requirement Understanding

The business goal of the system is to provide customers with a simple, reliable, and secure way to discover and buy books online, while giving the business owner tools to manage inventory, orders, and users. At a high level, the platform must support the following core workflows:

- A visitor can create an account and log in securely.
- A logged-in user can browse the book catalog by category or view featured/new releases.
- A user can search for books by title, author, or category and view detailed book information (price, description, availability, reviews).
- A user can add books to a shopping cart, update quantities, and remove items.
- A user can check out, enter shipping/payment details, and complete a purchase.
- The system confirms the order and allows the user to view order history and status.
- A user can manage their account details (profile, password, addresses).

These workflows form the basis for the functional requirements, user stories, and test scenarios below.

3. Functional Requirements

Functional requirements describe specific behaviours the system must support:

1. **FR1** - The system shall allow a new user to register using name, email, and password, with email verification.
2. **FR2** - The system shall allow a registered user to log in using valid credentials and log out at any time.
3. **FR3** - The system shall allow users to browse the book catalog by category, author, or new arrivals.
4. **FR4** - The system shall allow users to search for books by title, author, or category and display matching results.
5. **FR5** - The system shall display detailed book information, including title, author, price, description, stock availability, and customer ratings.
6. **FR6** - The system shall allow users to add one or more books to a shopping cart and view the cart contents.
7. **FR7** - The system shall allow users to update quantities or remove items from the shopping cart.
8. **FR8** - The system shall allow users to check out, provide shipping details, and complete payment through a supported payment method.
9. **FR9** - The system shall generate an order confirmation with an order number and estimated delivery date after successful payment.

10. FR10 - The system shall allow users to view their past orders and current order status from their account.

4. Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system:

1. **NFR1** - Performance: Catalog and search pages shall load within 3 seconds under normal load conditions.
2. **NFR2** - Security: User passwords and payment information shall be encrypted in transit and at rest, and the checkout process shall use a secure (HTTPS) connection.
3. **NFR3** - Availability: The platform shall be available 99.5% of the time, excluding scheduled maintenance windows.
4. **NFR4** - Usability: The interface shall be intuitive enough for a first-time user to complete a purchase without external help, and shall be accessible on both desktop and mobile browsers.
5. **NFR5** - Scalability: The system shall support at least a 5x increase in concurrent users during peak sales events (e.g., seasonal discounts) without significant degradation in performance.

5. User Stories

User stories describe system interactions from the perspective of different users:

1. **As a new visitor**, I want to register an account so that I can save my details and make purchases.
2. **As a registered user**, I want to log in securely so that I can access my account and order history.
3. **As a shopper**, I want to browse books by category so that I can discover titles that interest me.
4. **As a shopper**, I want to search for a book by title or author so that I can quickly find a specific book.
5. **As a shopper**, I want to view detailed information about a book so that I can decide whether to purchase it.
6. **As a shopper**, I want to add books to my cart so that I can purchase multiple items in one transaction.
7. **As a shopper**, I want to remove or update items in my cart so that I only pay for what I intend to buy.
8. **As a shopper**, I want to check out and pay securely so that I can complete my purchase with confidence.
9. **As a customer**, I want to receive an order confirmation so that I know my purchase was successful.
10. **As a customer**, I want to view my order history so that I can track past and current orders.

6. Acceptance Criteria (Gherkin Syntax)

The following scenarios define the acceptance criteria for key user stories, written using Given-When-Then syntax:

Scenario: User registration with valid details

Given a new visitor is on the registration page

When the visitor enters a valid name, email, and password and submits the form

Then the system creates a new account and sends a verification email

Scenario: Successful login

Given a registered user is on the login page

When the user enters a valid email and password

Then the system logs the user in and redirects to the homepage

Scenario: Search for a book

Given the user is on the bookstore homepage

When the user searches for a book by title

Then the system should display relevant search results

Scenario: View book details

Given the user has performed a search or is browsing the catalog
When the user selects a book from the results
Then the system displays the book's title, author, price, description, and availability

Scenario: Add a book to the cart

Given the user is viewing a book's details page and the book is in stock
When the user clicks "Add to Cart"
Then the book should be added to the cart and the cart item count should increase by one

Scenario: Remove a book from the cart

Given the user has at least one book in their cart
When the user clicks "Remove" on a cart item
Then the item should be removed from the cart and the total price should update accordingly

Scenario: Successful checkout and payment

Given the user has at least one book in the cart and has entered valid shipping and payment details
When the user submits the order
Then the payment should be processed successfully
And the system should display an order confirmation with an order number

7. Sprint Planning (High-Level)

The requirements above can be grouped into the following high-level sprints for iterative development and testing:

Sprint	Focus Area	Key Deliverables
Sprint 1	Account & Access	User registration, login/authentication, account management
Sprint 2	Catalog Browsing	Browse catalog, search by title/author/category, view book details
Sprint 3	Cart & Checkout	Add/remove cart items, checkout flow, payment processing
Sprint 4	Order & Polish	Order confirmation, order history, performance & security hardening

8. Conclusion

This document captures the functional requirements, non-functional requirements, user stories, and Gherkin-based acceptance criteria for the Online Book Store system. It provides a foundation for writing manual test cases and validating that the delivered system meets business expectations.